

Supplementary Material for Online Slip Detection and Friction Coefficient Estimation

I. SLIP DETECTION DELAY AND SAFETY IMPLICATIONS

To show the practical impact and safety implications of our approach's detection delay, we must contextualize them with respect to the vehicle's velocity and the average human reaction time. Our average slip detection delays are 0.396 ± 0.528 s for tile, 0.452 ± 0.489 s for cardboard, and 0.529 ± 0.610 s for acrylic. At the standard operating velocity of roughly 0.8 m/s, an average delay of 0.5s translates to a travel distance of approximately 0.4m before the slip is identified. Although the variance indicates that detection times occasionally exceed 1 second, the average detection times across all surfaces remain considerably lower than the average human reaction time (≥ 0.75 s). Thus, the system typically detects slip events at a rate comparable to or faster than a human driver.

II. FRICTION DETECTION ACCURACY AND LIMITATIONS

A. Variability in Friction-Coefficient Estimates

Although the average estimated friction coefficients were reasonably close to the measured ground-truth values, the individual estimates exhibited noticeable variability. This variance introduces an important limitation of the current approach. The variation in the estimations could have arisen from surface wear, tire condition, and the specific maneuver performed to generate a slip during each trial. The results shown support the method's ability to distinguish between surfaces with tire-road friction coefficients; however, testing on additional surfaces and using a more controlled procedure to induce consistent slip across trials would further strengthen its generalizability.

B. Interpreting the Friction Estimation Error

We report the estimation error of our approach in Section 5 (Experimental Evaluation). There is no reference value in the literature that defines how small this error must be. No standard (e.g., ISO or SAE) specifies a maximum allowable error for TRFC estimation, and the surveys and estimation papers we reviewed do not report one. For this reason, the error cannot be judged against a fixed target. Prior works instead evaluate their accuracy by comparing against existing methods, and we follow the same practice: Table 1 reports our error alongside established approaches, where our results fall in the same and comparable range while using fewer sensors and a lightweight model.

III. SUPPLEMENTAL FIGURES

Figure 1 shows an example of our TRFC estimation results on cardboard, acrylic, and tile surfaces. The red circle represents the friction circle, indicating the limit of possible tire force combinations where maximum friction is estimated. The variability of acceleration demonstrates that our detection and estimation approach was tested comprehensively.

Figure 2 shows the general range of velocity in m/s that the cars travel during the testing sessions.

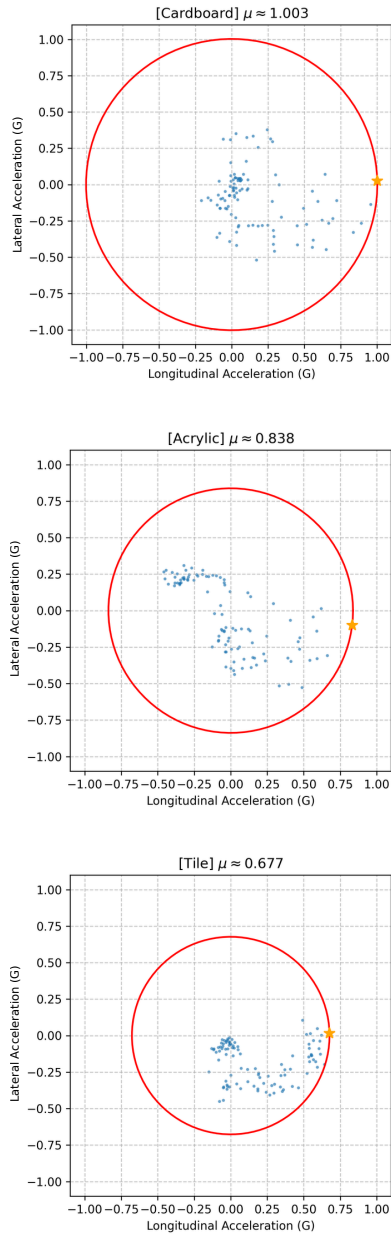


Fig. 1: Friction-circle visualizations for cardboard, acrylic, and tile surfaces. The blue dots denote measured longitudinal and lateral accelerations in units of g , while the star indicates the maximum traction point corresponding to the estimated friction coefficient μ .

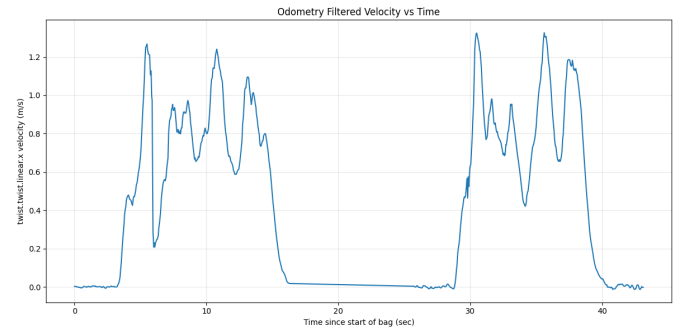


Fig. 2: Measured vehicle velocity over time during a representative experimental run.